Adaptive Pneumatic-Lead-Screw Automated Cutting Platform

About the Project

We first developed a soft robotic system—including a pneumatic gripper, cutting arm, and wiper—to automate vegetable slicing for users with fine motor disabilities, using silicone gripper, an ESP32 controller, and sensor feedback.

Midway, we pivoted to create a lab specimen cutter for wood and metal. The new design replaced kitchen-focused parts with a motorized cutting head on a horizontal lead screw, enabling precise, programmable cuts. Key components were 3D-printed for prototyping; the only metal element is the lead screw for load-bearing and precision motion. Both versions feature modular parts, easy-to-use controls, and prioritize safety.

Individual Contribution

Seershika :Electronics & Coding

Nalin :Cutting head Mechanism & 3D Printing

Kalpesh :Lead Screw Mechanism & CAD Modeling

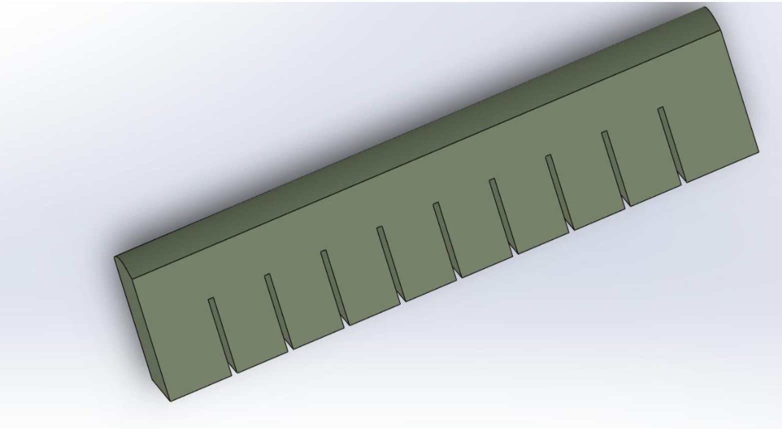
Priyanshu :Lead Screw Mechanism & CAD Modeling

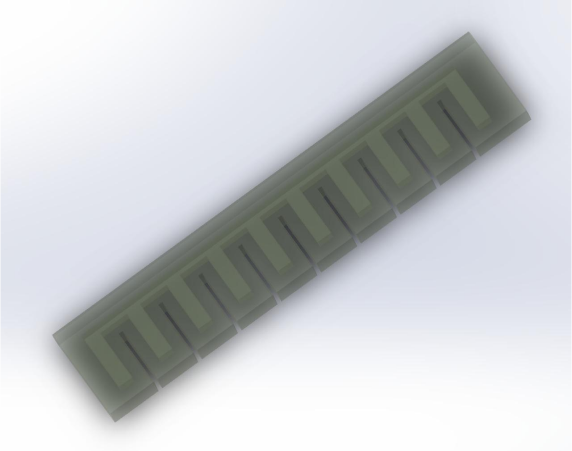
Ram Nivash:Electronics & Coding

Earlier Iterations

Soft Robotic Gripper

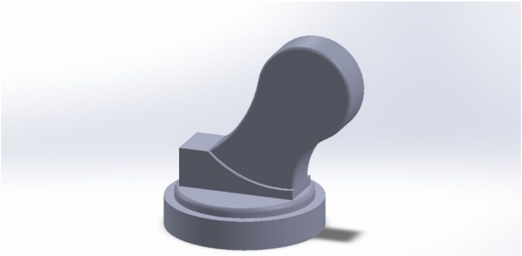
The soft robotic gripper was initially conceptualized in SolidWorks to safely handle delicate food items, with flexible slots enabling controlled bending via pneumatic or motor-driven actuation. However, due to change in concept of the overall project, this component was no longer required and was not fabricated. The modeling process still provided valuable insights into soft robotics and adaptive gripping challenges.

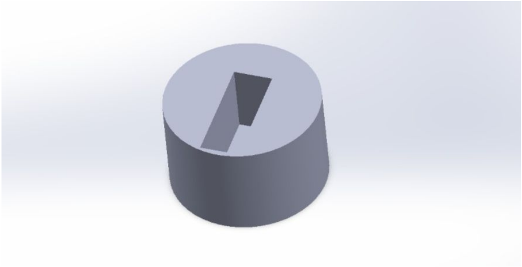


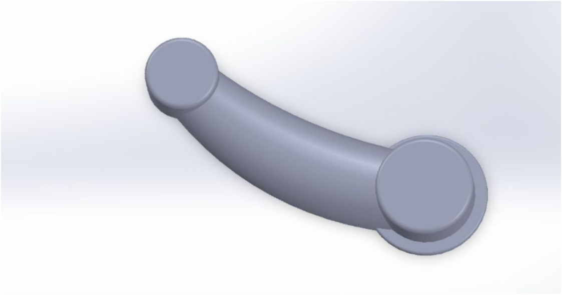


Robotic Arm

The robotic arm was designed as a 4-DOF system in SolidWorks to perform accurate food-cutting tasks, with a focus on flexible and smooth movement. It was not built because some components were not available. The arm was planned to work with a soft gripper so that fruits and vegetables could be cut easily, helping us better understand assistive robotic systems.



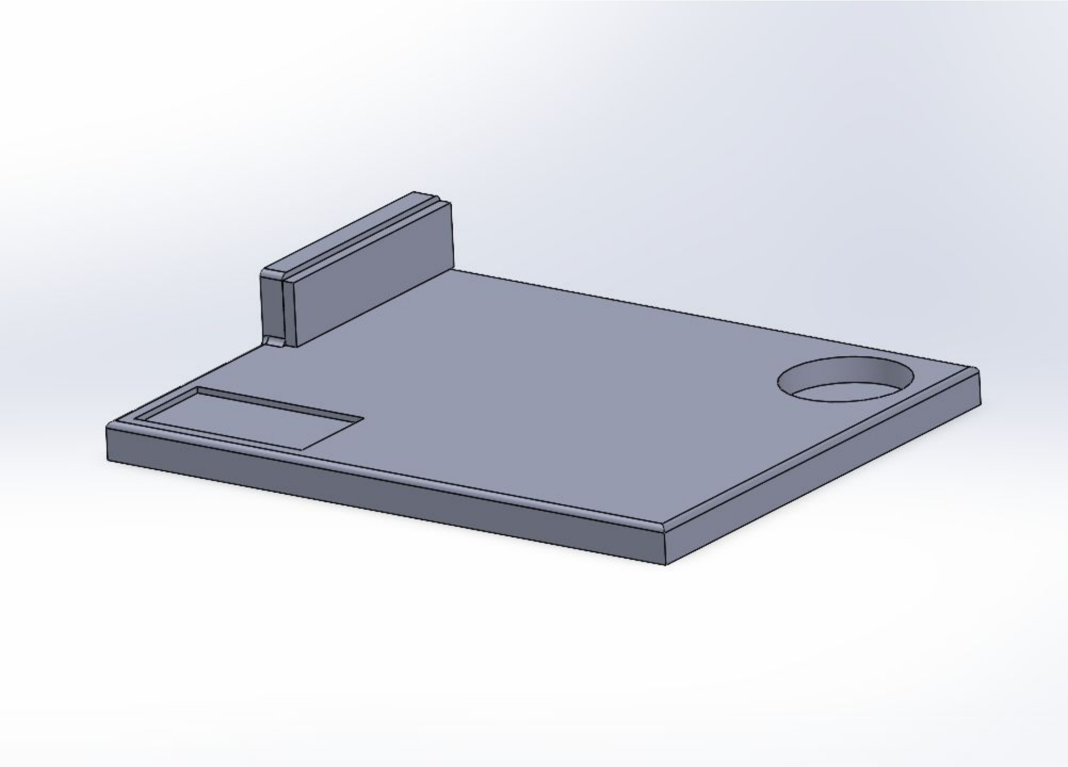




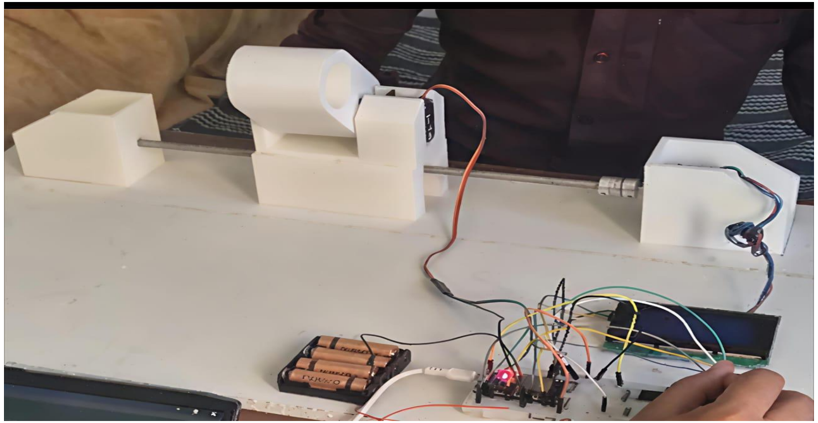
Mainframe

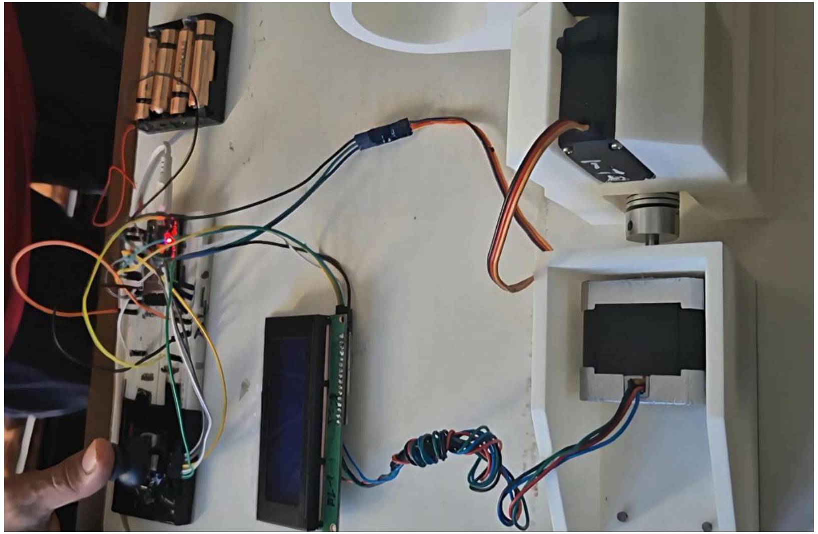
This would be the base of our device. It is equipped with a slider that can be used to swipe off the cut food items onto a plate automatically after the cutting process has finished.

All the controlling systems like joystick and buttons would also be placed on this platform.



Hardware Implementation

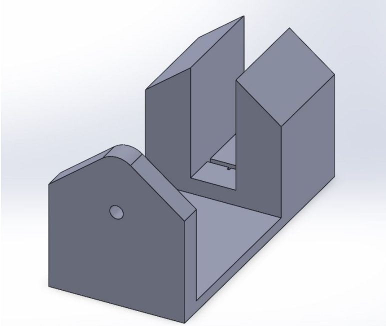


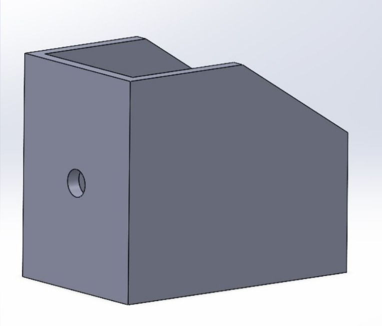


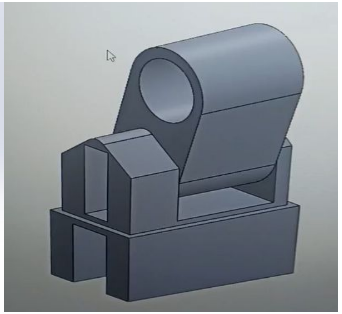
Current Design of the Movement Mechanism

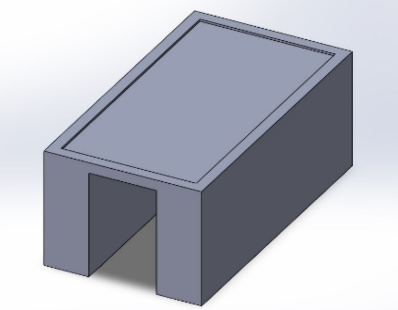
Lead Screw Mechanism

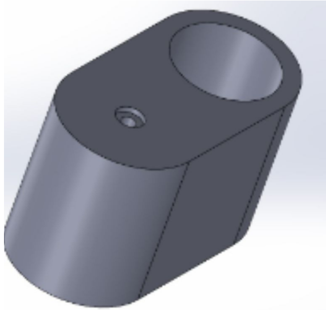
As initial Design was based on 6 servo motor, and due to lack of motors we make some changes in the design.So in this design we only need 1 servo and 2 stepper motors.In this mechanism we use threaded screw to convert rotary motion of stepper motor into linear motion. This design is designed as 2-DOF system to perform parallel and specific distance cutting By blade which is attached to arm.

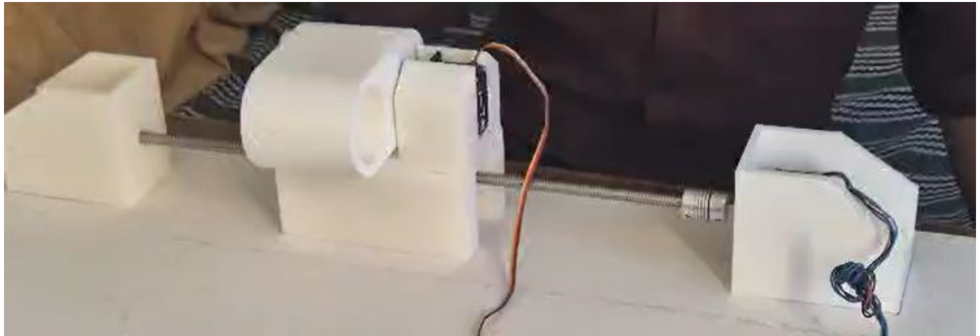


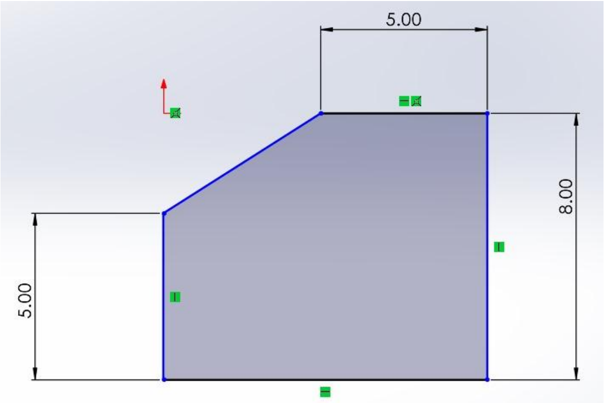


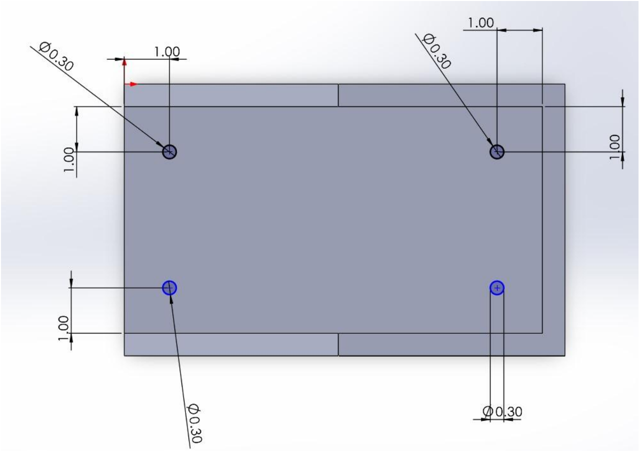




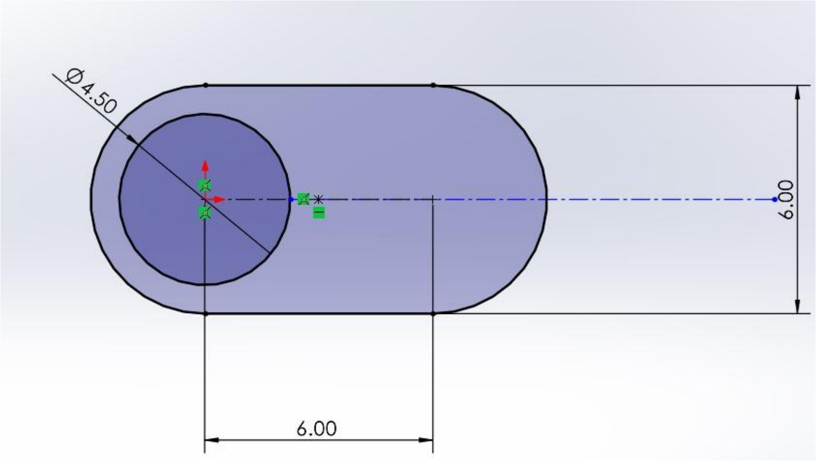


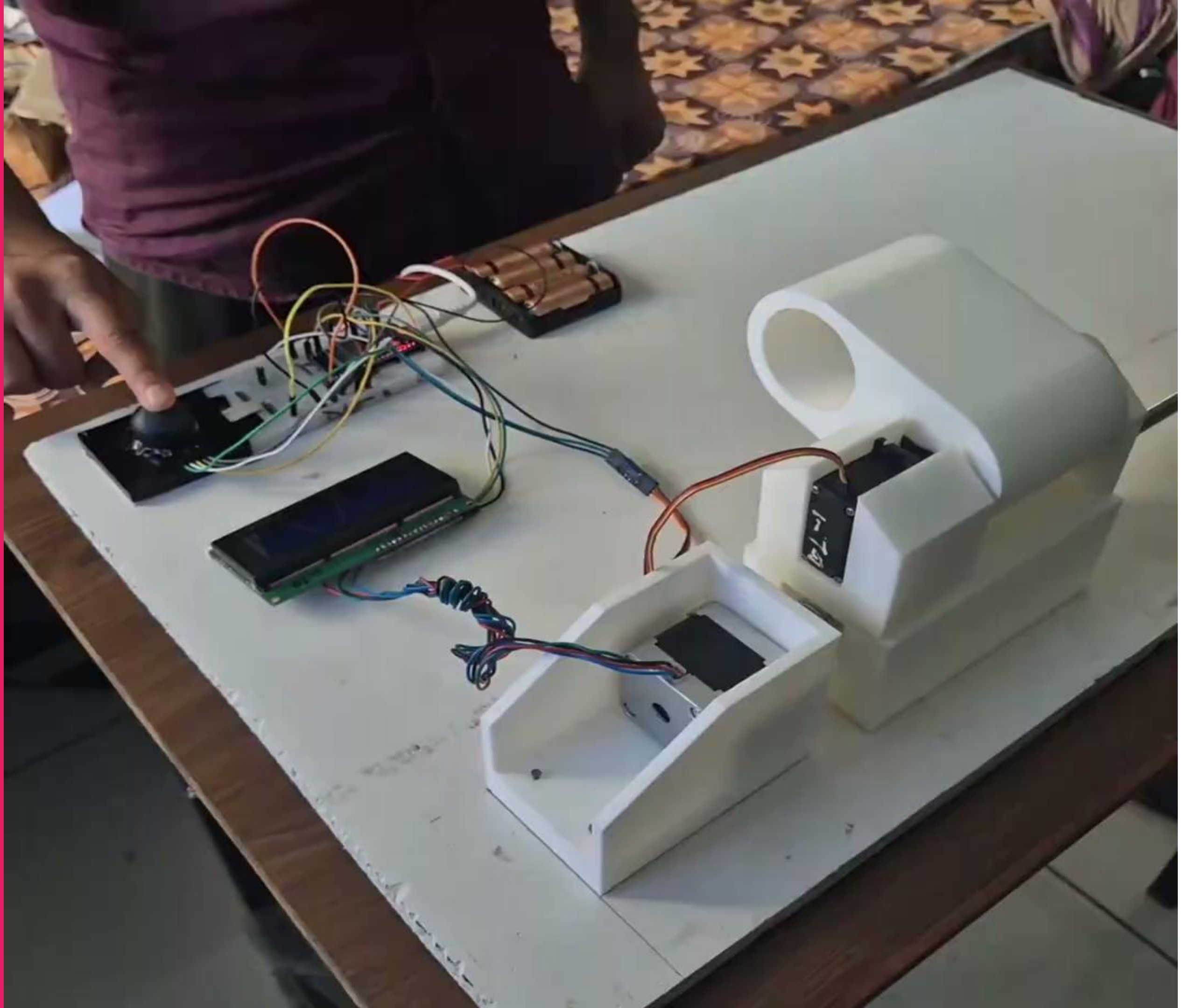




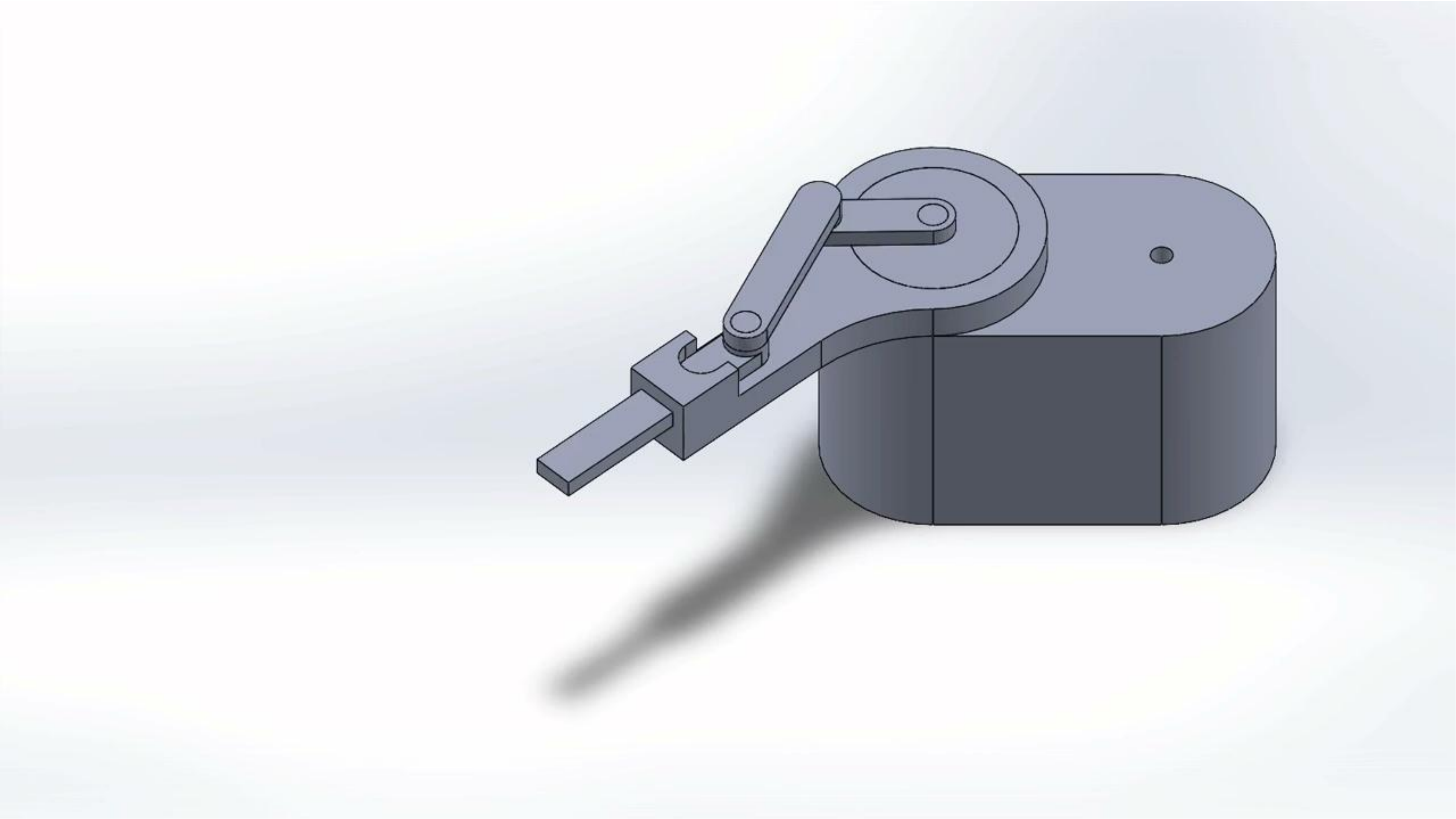


To set all the components on the above page we used the dimension given beside for clamp and arm. Also we used the proper dimension for base set on the threaded rod so that stepper motor can handle the weight.We also build holes for the motor with proper dimension of motors.





Current Design of the Cutting Mechanism



Design Details

Crank Length (r) = 3cm

Coupler Length (l) = 5cm

Slider Length = 7cm

Stroke Length = 6cm

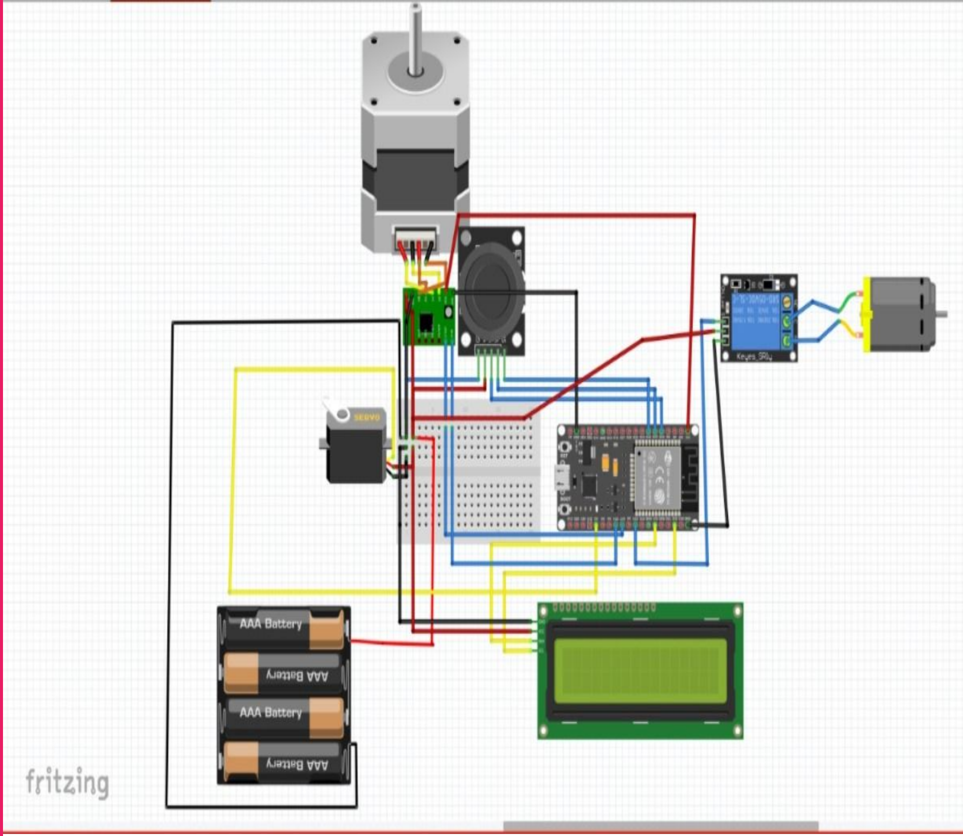
Minimum Torque required for the DC Motor = 3.54Nm

The Electronics

The Electronics Used :

1. Servo MG995R
2. Stepper Motor NEMA17+A4988 Stepper Motor Driver
3. High Torque DC Motor + 1 Mechanical Relay
4. ESP32
5. Joystick Module
6. I2C Display

Circuit diagram



We have used fritzing to plan out connections and test them.

Thank You

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